TS. Please:
- Change all words underlined in BLUE to have initial cap
- Lowercase all words underlined in GREEN
- Number Examples, Theorems, etc. decimally by chapter in a single sequence, e.g. Example 3.1, Definition 3.2, Theorem 3.3, Remark 3.4 etc.
- Examples/Definitions/Claims/Remarks should be set in roman. Propositions/Theorems/Corollaries/Lemmas should remain italicised thoughout. Please amend.

~~Chapter~~ 4

Markov Chains

Our goal in extending the deterministic decision making framework of Chapter 3 to stochastic models is to develop a mathematical model for decision making under uncertainty. With the goal of presenting such a model in mind, in this chapter we cover the fundamentals of the Markov chain stochastic process. A detailed treatment can be found in various books, e.g., [96, 121].

Chapter Orientation

Main Prerequisites:

- Probability theory

Main terminology:

- $X_{\mathtt{t}}$ state of the Markov chain at time $\mathtt{t}$
- $p_{i,j}$ probability of transitioning from state i to state j
- $p_{i,j}^{(m)}$ m-step transition probabilities
- P transition matrix
- p_0 initial state distribution
- d_i period of state i
- T_i return time to state i

- μ invariant distribution

A *Markov chain* $\{X_{\mathtt{t}},\ \mathtt{t} = 0, 1, 2, \ldots\}$, with $X_{\mathtt{t}} \in X$, is a discrete-time stochastic process, over a finite or countable state-space X, that satisfies the following Markov property:

$$\Pr(X_{\mathtt{t}+1} = j | X_{\mathtt{t}} = i, X_{\mathtt{t}-1}, \ldots X_0) = \Pr(X_{\mathtt{t}+1} = j | X_{\mathtt{t}} = i).$$

We focus on time-homogeneous Markov chains, where

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$$\Pr(X_{\mathtt{t}+1} = j | X_{\mathtt{t}} = i) = \Pr(X_1 = j | X_0 = i) \triangleq p_{i,j}.$$

The $p_{i,j}$'s are the transition probabilities, which satisfy $p_{i,j} \geq 0$, and for each $i \in X$ we have $\sum_{j \in X} p_{i,j} = 1$, namely, $\{p_{i,j} : j \in X\}$ is a distribution on the next state following state i. The matrix $P = (p_{i,j})$ is the transition matrix. The matrix is row-stochastic (each row sums to 1 and all entries non-negative).

Given the initial distribution p_0 of X_0, namely $\Pr(X_0 = i) = p_0(i)$, we obtain the finite-dimensional distributions for states $i_0, i_1, \ldots, i_t$:

$$\Pr(X_0 = i_0, X_1 = i_1, \ldots, X_{\mathtt{t}} = i_{\mathtt{t}}) = p_0(i_0) p_{i_0,i_1} \cdot \ldots \cdot p_{i_{\mathtt{t}-1},i_{\mathtt{t}}}.$$

Define $p_{i,j}^{(m)} = \Pr(X_m = j | X_0 = i)$, the m-step transition probabilities. It is easy to verify that $p_{i,j}^{(m)} = [P^m]_{ij}$, where P^m is the m-th power of the matrix P.

Example 4.1. *Consider the following two state Markov chain, with transition probability P and initial distribution p_0, as follows:*

$$P = \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix} \qquad p_0 = \begin{pmatrix} 0.5 & 0.5 \end{pmatrix}$$

Initially, we have both states equally likely. After one step, the distribution of states is $p_1 = p_0 P = (0.3\ ,\ 0.7)$. After two steps we have $p_2 = p_1 P = p_0 P^2 = (0.26\ ,\ 0.74)$. The limit of this sequence would be $p_\infty = (0.25\ ,\ 0.75)$, which is called the steady state distribution, *and would be discussed later.*

4.1 State Classification

We classify the Markov chain given the ability to transition between states. We start by defining the *accessibility* (or *reachability*) relation between two states, which capture the fact that the Markov state can transition from one state to another with some positive probability.

Definition 4.1. *State j is* accessible *(or* reachable*) from i (denoted by $i \to j$) if $p_{i,j}^{(m)} > 0$ for some $m \geq 1$.*

For a finite state space X we can compute the accessibility property as follows. Construct a directed graph $G(X, E)$ where the vertices are the states X and there is a directed edge (i, j) if $p_{i,j} > 0$. State j is accessible from state i iff there exists a directed path in $G(X, E)$ from i to j.
Note that the relation is transitive. If $i \to j$ and $j \to k$ then $i \to k$. This follows since $i \to j$ implies that there is m_1 such that $p_{i,j}^{(m_1)} > 0$. Similarly, since $j \to k$ there is m_2 such that $p_{j,k}^{(m_2)} > 0$. Therefore, for $m = m_1 + m_2$ we have $p_{i,k}^{(m)} > p_{i,j}^{(m_1)} p_{j,k}^{(m_2)} > 0$. States i and j are *communicating* states (or communicate) if $i \to j$ and $j \to i$. For a finite X, this implies that in $G(X, E)$ there is both a directed path from i to j and from j to i.

Definition 4.2. *A* communicating class *(or just* class*) is a maximal collection of states that communicate.*

For a finite state space X, this implies that in $G(X, E)$ we have i and j in the same strongly connected component of the graph. (A strongly connected component has a directed path between any pair of vertices.)

Definition 4.3. *The Markov chain is* irreducible *if all states belong to a single class (i.e., all states communicate with each other).*

For a finite state space X, this implies that $G(X, E)$ is strongly connected.

Next, we define the period of a state in a Markov chain. The period intuitively captures the fact that the time between any two occurrences of the state is always divisible by some integer, which is the period.

Definition 4.4. *State i has ~~a~~* period *$d_i = GCD\{m \geq 1 : p_{i,i}^{(m)} > 0\}$, where GCD is the greatest common divisor. A state is* aperiodic *if $d_i = 1$.*

State i is periodic with period $d_i \geq 2$ if $p_{i,i}^{(m)} = 0$ for $m \pmod{d_i} \neq 0$ and for any m such that $p_{i,i}^{(m)} > 0$ we have $m \pmod{d_i} = 0$. If a state i is aperiodic, then there exists an integer m_0 such that for any $m \geq m_0$ we have $p_{i,i}^{(m)} > 0$. Periodicity is a class property: all states in the same class have the same period. Specifically, if some state is *aperiodic*, then all states in the class are aperiodic.

Claim 4.1. *For any two states i and j with periods d_i and d_j, in the same communicating class, we have $d_i = d_j$.*

Proof. For contradiction, assume that $d_j \pmod{d_i} \neq 0$. Since they are in the same communicating class, we have a trajectory from i to j of length $m_{i,j}$ and from j to i of length $m_{j,i}$. This implies that $(m_{i,j} + m_{j,i}) \pmod{d_i} = 0$. Now, there is a trajectory (which is a cycle) of length $m_{j,j}$ from j back to j such that $m_{j,j} \pmod{d_i} \neq 0$ (otherwise d_i divides the period of j). Consider the path from i to itself of length $m_{i,j}+m_{j,j}+m_{j,i}$. We have that $(m_{ij}+m_{jj}+m_{ji}) \pmod{d_i} = m_{jj} \pmod{d_i} \neq 0$. This is a contradiction to the definition of d_i. Therefore, $d_j \pmod{d_i} = 0$ and similarly $d_i \pmod{d_j} = 0$, which implies that $d_i = d_j$. □

The claim shows that periodicity is a class property, and all the states in a class have the same period.

4.2 Recurrence

We define the property that a state will reappear in the Markov chain. A *recurrent state* is a state which with probability 1 will reappear in the Markov chain, otherwise it is a *transient state*. The formal definition is following.

Definition 4.5. *State i is* recurrent *if* $\Pr(X_{\mathtt{t}} = i$ *for some* $\mathtt{t} \geq 1 | X_0 = i) = 1$. *Otherwise, state i is* transient.

We can relate the state property of recurrent and transient to the expected number of returns to a state.

Claim 4.2. *State i is transient iff* $\sum_{m=1}^{\infty} p_{i,i}^{(m)} < \infty$.

Proof. Assume that state i is transient. Let $q_i = \Pr(X_{\mathtt{t}} = i$ for some $\mathtt{t} \geq 1 | X_0 = i)$. Since state i is transient we have $q_i < 1$. Let Z_i be the number of times the trajectory returns to state i. Note that Z_i is geometrically distributed with parameter q_i, namely $\Pr[Z_i = k] = q_i^k(1-q_i)$. Therefore the expected number of returns to state i is $\boxed{\mathbb{E}[Z] = \sum_{k=0}^{\infty} k(1-q_i)p^k = (1-q_i)\sum_{k=0}^{\infty} kq_i^k = (1-q_i)\frac{q_i}{(1-q_i)^2} = \frac{q_i}{1-q_i}}$ and is finite. The expected number of returns to state i is equivalently $\sum_{m=1}^{\infty} p_{i,i}^{(m)}$ and hence if a state is transient we have $\sum_{m=1}^{\infty} p_{i,i}^{(m)} < \infty$.

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For the other direction, assume that $\sum_{m=1}^{\infty} p_{i,i}^{(m)} < \infty$. This implies that there is an m_0 such that $\sum_{m=m_0}^{\infty} p_{i,i}^{(m)} < 1/2$. Consider the probability of returning to i within m_0 steps. This implies that $\Pr(X_{\mathtt{t}} = i$ for some $t \geq m_0 | X_0 = i) < 1/2$. Now consider the probability $q_i' = \Pr(X_{\mathtt{t}} = i$ for some $m_0 \geq \mathtt{t} \geq 1 | X_0 = i)$. If $q_i' < 1$, this implies that $\Pr(X_{\mathtt{t}} = i$ for some $\mathtt{t} \geq 1 | X_0 = i) < q_i' + (1-q_i')/2 = (1+q_i')/2 < 1$, which implies that state i is transient. If $q_i' = 1$, this implies that after at most m_0

that is,

steps we are guaranteed to return to i, hence the expected number of return to state i is infinite, i.e., $\sum_{m=1}^{\infty} p_{i,i}^{(m)} = \infty$. This is in contradiction to the assumption that $\sum_{m=1}^{\infty} p_{i,i}^{(m)} < \infty$. □

Claim 4.3. *Recurrence is a class property.*

Proof. To see this consider two states i and j in the same communicating class and i is recurrent. Since i is recurrent $\sum_{m=1}^{\infty} p_{i,i}^{(m)} = \infty$. Since j is accessible from i there is a k such that $p_{i,j}^{(k)} > 0$. Since i is accessible from j, there exists a k' such that $p_{j,i}^{(k')} > 0$. We can lower bound $\sum_{m=1}^{\infty} p_{j,j}^{(m)}$ by $\sum_{m=1}^{\infty} p_{j,j}^{(m)} \geq p_{j,i}^{(k')} \left(\sum_{m=1}^{\infty} p_{i,i}^{(m)}\right) p_{i,j}^{(k)} = \infty$. Therefore we showed that state j is recurrent. □

Claim 4.4. *If states i and j are in the same recurrent (communicating) class, then state j is (eventually) reached from state i with probability 1, namely,* $\Pr(X_{\mathtt{t}} = j$ *for some* $t \geq 1 | X_0 = i) = 1$.

Proof. This follows from the fact that both states occur infinitely often, with probability 1. □

Definition 4.6. *Let T_i be the* first return time *to state i, defined as the minimum $t \geq 1$ such that $X_t = i$, given $X_0 = i$. That is, $T_i = \inf\{t \geq 1 : X_t = i\}$.*

Claim 4.5. *If i is a recurrent state, then $T_i < \infty$ w.p. 1.*

Proof. Since otherwise, there is a positive probability that we never return to state i, and hence state i is not recurrent. □

We now would like consider the expected time to return to a state. The claim shows that any recurrent state has a finite return time, but this does not imply that the expected return time is finite. (For example, consider a random variable T such that $\Pr[T = 2^k] = 2^{-k}$. In this example, $\mathbb{E}[T] = \sum_k 2^k 2^{-k} = \sum_k 1 = \infty$.)

Definition 4.7. *State i is* positive recurrent *if $\mathbb{E}(T_i) < \infty$, and null recurrent if $\mathbb{E}(T_i) = \infty$.*

We now show that in the case of finite state space, we do not have null recurrent states.

Claim 4.6. *If the state space X is finite, all recurrent states are positive recurrent.*

Proof. This follows since the set of states that are null recurrent cannot have transitions from positive recurrent states and cannot have a transition to transient states. If the chain never leaves the set of null recurrent states, then some state would have a return time which is at most the size of the set. If there is a positive probability of leaving the set (and never returning) then the states are transient. (See the proof of Theorem 4.10 for a more formal proof of a similar claim for countable Markov Chains.) □

In the following we illustrate some of the notions that we define. we start with the classic random walk on the integers, where all integer (states) are null recurrent.

Example 4.2. **Random walk** *Consider the following Markov chain over the integers. The states are the integers. The initial state is* 0*. At each state* i*, with probability* $\frac{1}{2}$ *we move to* $i+1$ *and with probability* $\frac{1}{2}$ *to* $i-1$*. Namely,* $p_{i,i+1} = \frac{1}{2}$*,* $p_{i,i-1} = \frac{1}{2}$*, and* $p_{i,j} = 0$ *for* $j \notin \{i-1, i+1\}$*. We will show that* T_i *is finite with probability* 1 *and* $\mathbb{E}[T_i] = \infty$*. This implies that all the states are null recurrent.*

To compute $\mathbb{E}[T_i]$ *consider what happens after one and two steps. Let* $Z_{i,j}$ *be the time to move from* i *to* j*. Note that we have*

$$\mathbb{E}[T_i] = 1 + 0.5\mathbb{E}[Z_{i+1,i}] + 0.5\mathbb{E}[Z_{i-1,i}] = 1 + \mathbb{E}[Z_{1,0}],$$

since, due to symmetry, $\mathbb{E}[Z_{i,i+1}] = \mathbb{E}[Z_{i+1,i}] = \mathbb{E}[Z_{1,0}]$*.*

After two steps we are either back to i*, or at state* $i+2$ *or state* $i-2$*. For* $\mathbb{E}[T_i]$ *we have that*

$$\begin{aligned}\mathbb{E}[T_i] =& 2 + \frac{1}{4}\mathbb{E}[Z_{i+2,i}] + \frac{1}{4}\mathbb{E}[Z_{i-2,i}] \\ =& 2 + \frac{1}{4}(\mathbb{E}[Z_{i+2,i+1}] + \mathbb{E}[Z_{i+1,i}]) + \frac{1}{4}(\mathbb{E}[Z_{i-2,i-1}] + \mathbb{E}[Z_{i-1,i}]) \\ =& 2 + E[Z_{1,0}],\end{aligned}$$

where the first identity uses the fact that $Z_{i+2,i} = Z_{i+2,i+1} + Z_{i+1,i}$*, since in order to reach from state* $i+2$ *to state* i *we need to first reach from state* $i+2$ *state* $i+1$*, and then from state* $i+1$ *to state* i*.*

This implies that we have

$$1 + \mathbb{E}[Z_{1,0}] = \mathbb{E}[T_i] = 2 + E[Z_{1,0}]$$

Clearly, there is no finite value for $\mathbb{E}[Z_{1,0}]$ *which will satisfy both equations, which implies* $\mathbb{E}[Z_{1,0}] = \infty$*, and hence* $\mathbb{E}[T_i] = \infty$*.*

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To show that state 0 is a recurrent state, note that the probability that at time $2k$ we are at state 0 is exactly $\boxed{p_{0,0}^{(2k)} = \binom{2k}{k} 2^{-2k} \approx \frac{c}{\sqrt{k}}}$ *(using Stirling's approximation), for some constant $c > 0$. This implies that*

$$\sum_{m=1}^{\infty} p_{0,0}^{(m)} \approx \sum_{m=1}^{\infty} \frac{c}{\sqrt{m}} = \infty$$

and therefore state 0 is recurrent. (By symmetry, this shows that all the states are recurrent.)

Note that this Markov chain has a period of 2. This follows since any trajectory starting at 0 and returning to 0 has an equal number of $+1$ and -1 and therefore of even length. Any even number n has a trajectory of this length that starts at 0 and returns to 0, for example, having $\frac{n}{2}$ times $+1$ followed by $\frac{n}{2}$ times -1.

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The next example is a simple modification of the random walk, where each time we either return to the origin or continue to the next integer with equal probability. This Markov chain will have all (non-negative) integers as positive recurrent states.

Example 4.3. Random walk with jumps. *Consider the following Markov chain over the integers. The states are the integers. The initial state is 0. At each state i, with probability $\frac{1}{2}$ we move to $i+1$ and with probability $\frac{1}{2}$ we return to 0. Namely, $p_{i,i+1} = \frac{1}{2}$, $p_{i,0} = \frac{1}{2}$, and $p_{i,j} = 0$ for $j \notin \{0, i+1\}$. We will show that $\mathbb{E}[T_i] < \infty$ (which implies that T_i is finite with probability 1).*

From any state we return to 0 with probability $\frac{1}{2}$, therefore $\mathbb{E}[T_0] = 2$ (The return time is 1 with probability $\frac{1}{2}$, 2 with probability $\left(\frac{1}{2}\right)^2$, k with probability $\left(\frac{1}{2}\right)^k$, and computing the expectation gives $\sum_{k=1}^{\infty} \frac{k}{2^k} = 2$). We will show that for state i we have $\mathbb{E}[T_i] \leq 2 + 2 \cdot 2^i$. We will decompose T_i to two parts. The first is the return to 0, this part has expectation 2. The second is to reach state i from state 0. Consider an epoch as the time between two visits to 0. The probability that an epoch would reach i is exactly 2^{-i}. The expected time of an epoch is 2 (the expected time to return to state 0). The expected time to return to state 0, given that we did not reach state i is less than 2. Therefore, $\mathbb{E}[T_i] \leq 2 + 2 \cdot 2^i$. Note that this Markov chain is aperiodic.

4.3 Invariant Distribution

In general, we have an expected state distribution of the Markov chain depending on the time and the start state. We would like to capture what happens as we run the Markov chain for a long time. Intuitively, we would like to see convergence of the state distribution. For this we need the Markov chain to be aperiodic, since otherwise

the periodicity will not allow convergence, since a periodic state will switch from zero to non-zero probability. We would also like to have the Markov chain irreducible, otherwise some states will have a vanishing but non-zero probability. This limit distribution is called in a few different names, all meaning the same concept: invariant distribution, stationary distribution, and steady state distribution.

The probability vector $\mu = (\mu_i)$ is an *invariant distribution* (or *stationary distribution* or *steady state distribution*) for the Markov chain if $\mu^\top P = \mu^\top$, namely

$$\mu_j = \sum_i \mu_i p_{i,j} \quad \forall j.$$

Clearly, if $X_\mathtt{t} \sim \mu$ then $X_{\mathtt{t}+1} \sim \mu$. If $X_0 \sim \mu$, then the Markov chain $(X_\mathtt{t})$ is a *stationary stochastic process.*

Theorem 4.7. *Let $(X_\mathtt{t})$ be an irreducible and aperiodic Markov chain over a finite state space X with transition matrix P. Then there is a unique distribution $\mu > 0$ such that $\mu^\top P = \mu^\top > 0$.*

Proof. Assume that x is an eigenvector of P with eigenvalue λ, i.e., we have $Px = \lambda x$. Since P is a stochastic matrix, we have $\|Px\|_\infty \leq \|x\|_\infty$, which implies that $|\lambda| \leq 1$. Since the matrix P is row stochastic, $P\vec{1} = \vec{1}$, which implies that P has a right eigenvalue of 1 and this is the maximal eigenvalue. Since the sets of right and left eigenvalues are identical for square matrices, we conclude that there is x such that $x^\top P = x^\top$. Our first task is to show that there is such an x with $x \geq 0$.

Since the Markov chain is irreducible and aperiodic, there is an integer m, such that P^m has all ~~the~~ entries strictly positive. Namely, for any $i, j \in X$ we have $p_{i,j}^{(m)} > 0$.

We now show a general property of positive matrices (matrices where all the entries are strictly positive). Let $A = P^m$ be a positive matrix and x an eigenvector of A with eigenvalue 1. First, if x has complex entries then $Re(x)$ and $Im(x)$ are eigenvectors of A of eigenvalue 1, and one of them is non-zero. Therefore we can assume that $x \in \mathbb{R}^d$. We would like to show that there is an $x \geq 0$ such that $x^\top A = x^\top$. If $x \geq 0$ we are done. If $x \leq 0$ we can take $x' = -x$ and we are done. We need to show that x cannot have both positive and negative entries.

For contradiction, assume that we have $x_k > 0$ and $x_{k'} < 0$. This implies that for any weight vector $w > 0$ we have $|x^\top w| < |x|^\top w$, where $|x|$ is point-wise absolute value. Therefore,

$$\sum_j |x_j| = \sum_j \Big|\sum_i x_i A_{i,j}\Big| < \sum_j \sum_i |x_i| A_{i,j} = \sum_i |x_i| \sum_j A_{i,j} = \sum_i |x_i|,$$

where the first identity follows since x is an eigenvector. The second since P is strictly positive. The third is a change of order of summation. The last follows since P is a row stochastic matrix, so each row sums to 1. Clearly, we reached a contradiction, and therefore x cannot have both positive and negative entries.

We have shown so far that there exists a μ such that $\mu^\top P = \mu^\top$ and $\mu \geq 0$. This implies that $\frac{\mu}{\|\mu\|_1}$ is a steady state distribution. Since $A = P^m$ is strictly positive, then $\mu^\top = \mu^\top A > 0$.

To show the uniqueness of μ, assume we have x and y such that $x^\top P = x^\top$ and $y^\top P = y^\top$ and $x \neq y$. Recall that we showed that in such a case both $x > 0$ and $y > 0$. Then there is a linear combination $z = ax + by$ such that for some i we have $z_i = 0$. Since $z^\top P = z^\top$, we have showed that z is strictly positive, i.e., $z > 0$, which is a contradiction. Therefore, $x = y$, and hence μ is unique. □

We define the average fraction that a state $j \in X$ occurs, given that we start with an initial state distribution x_0 up to time m, as follows:

$$\pi_j^{(m)} = \frac{1}{m} \sum_{t=1}^{m} \mathbb{I}(X_t = j).$$

Theorem 4.8. *Let (X_t) be an irreducible and aperiodic Markov chain over a finite state space X with transition matrix P. Let μ be the stationary distribution of P. Then, for any $j \in X$ we have,*

$$\mu_j = \lim_{m\to\infty} \mathbb{E}[\pi_j^{(m)}] = \frac{1}{\mathbb{E}[T_j]}.$$

Proof. We have that

$$\mathbb{E}[\pi_j^{(m)}] = \mathbb{E}\Big[\frac{1}{m}\sum_{t=1}^{m} \mathbb{I}(X_t = j)\Big] = \frac{1}{m}\sum_{t=1}^{m} \Pr[X_t = j | X_0 = x_0] = \frac{1}{m}\sum_{t=1}^{m} x_0^\top P^t e_j,$$

where e_j denotes a vector of zeros, with 1 only in the j's element. Let $v_1, \ldots, v_n$ be the eigenvectors of P with eigenvalues $\lambda_1 \geq \ldots \geq \lambda_n$. By Theorem 4.7 we have that $v_1 = \mu$, the stationary distribution and $\lambda_1 = 1 > \lambda_i$ for $i \geq 2$. Rewrite $x_0 = \sum_i \alpha_i v_i$. Since P^m is a stochastic matrix, $x_0^\top P^m$ is a distribution, and therefore $\lim_{m\to\infty} x_0^\top P^m = \mu$ (this is known as the limit law of Markov Chains. For a proof, see [69], Theorem 1.3 on p.85-86).

We will be interested in the limit $\pi_j = \lim_{m\to\infty} \pi_j^{(m)}$, and mainly in the expected value $\mathbb{E}[\pi_j]$. From the above we have that $\mathbb{E}[\pi_j] = \mu_j$.

A different way to express $\mathbb{E}[\pi_j]$ is using a variable time horizon, with a fixed number of occurrences of j. Let $T_{k,j}$ be the time between the k and $k+1$ occurrence of state j. This implies that

$$\lim_{m\to\infty} \frac{1}{m} \sum_{t=1}^{m} \mathbb{I}(X_t = j) = \lim_{n\to\infty} \frac{n}{\sum_{k=1}^{n} T_{k,j}}.$$

Note that the $T_{k,j}$ are i.i.d. because of the Markov assumption and equivalent to T_j. By the law of large numbers we have that $\frac{1}{n}\sum_{k=1}^{n} T_{k,j}$ converges to $\mathbb{E}[T_j]$. Therefore,

$$\mathbb{E}[\pi_j] = \frac{1}{\mathbb{E}[T_j]}.$$

□

We have established the following general theorem.

Theorem 4.9 (**Recurrence of finite Markov chains**)**.** *Let $(X_\mathtt{t})$ be an irreducible, aperiodic Markov chain over a finite state space X. Then the following properties hold:*

1. *All states are **positive recurrent***
2. *There exists a **unique stationary distribution** μ, where $\mu_i = \mathbb{E}[T_i]^{-1}$.*
3. ***Convergence** to the stationary distribution:* $\lim_{\mathtt{t}\to\infty} \Pr[X_\mathtt{t} = j] = \mu_j \quad (\forall j)$
4. ***Ergodicity**: For any finite f:* $\lim_{\mathtt{t}\to\infty} \frac{1}{t}\sum_{s=0}^{\mathtt{t}-1} f(X_s) = \sum_i \mu_i f(i) \triangleq \pi \cdot f$.

We prove this theorem for the case that P is a diagonizable matrix. In the proof we sketch how to extend this result to the general case.

Proof. From Theorem 4.7, we have that $\mu > 0$, and from Theorem 4.8 we have that $\mathbb{E}[T_i] = \mu_i^{-1} < \infty$. This establishes (1) and (2).

For any initial distribution x_0 we have that

$$\Pr[X_t = j] = x_0^\top P^t e_j,$$

where e_j denotes a vector of zeros, with 1 only in the j's element. Let $v_1, \ldots, v_n$ be the eigenvectors of P with eigenvalues $\lambda_1 \geq \ldots \geq \lambda_n$. By Theorem 4.7 we have that

$v_1 = \mu$, the stationary distribution and $\lambda_1 = 1 > \lambda_i$ for $i \geq 2$. Assume that P is diagonizable, and rewrite $x_0 = \sum_i \alpha_i v_i$. [1] We have that

$$\Pr[X_{\mathtt{t}} = j] = \sum_i \alpha_i \lambda_i^t v_i^\top e_j,$$

and therefore $\lim_{\mathtt{t}\to\infty} \Pr[X_{\mathtt{t}} = j] = \lambda_1 \mu^\top e_j = \lambda_1 \mu_j$. Since P^t is a stochastic matrix, $x_0^\top P^t$ is a distribution, and therefore $\lambda_1 = 1$. This establishes (3).

Finally, we establish (4) following the proof of Theorem 4.8:

$$\begin{aligned}
\lim_{\mathtt{t}\to\infty} \frac{1}{\mathtt{t}} \sum\nolimits_{s=0}^{\mathtt{t}-1} f(X_s) &= \lim_{\mathtt{t}\to\infty} \frac{1}{\mathtt{t}} \sum\nolimits_{s=0}^{\mathtt{t}-1} \sum\nolimits_i \mathbb{I}(X_s = i) f(i) \\
&= \sum\nolimits_i f(i) \lim_{\mathtt{t}\to\infty} \frac{1}{\mathtt{t}} \sum\nolimits_{s=0}^{\mathtt{t}-1} \mathbb{I}(X_s = i) \\
&= \sum\nolimits_i \mu_i f(i).
\end{aligned} \tag{4.1}$$

□

For countable Markov chains, unlike finite Markov chains, there are additional possibilities beyond recurrence.

Theorem 4.10 (**Countable Markov chains**)**.** *Let $(X_{\mathtt{t}})$ be an irreducible Markov chain over a countable state space X. Then: Either (i) all states are positive recurrent, or (ii) all states are null recurrent, or (iii) all states are transient.*

Proof. Let i be a positive recurrent state. We will show that all states are positive recurrent. For any state j, since the Markov chain is irreducible, we have for some $m_1, m_2 \geq 0$ that $p_{j,i}^{(m_1)}, p_{i,j}^{(m_2)} > 0$. This implies that the expected return time to state j is upper bounded by the returning to i, transitionning from j to i and from i to j. That is:

$$\mathbb{E}[T_j] \leq \frac{1}{p_{j,i}^{(m_1)}} + \mathbb{E}[T_i] + \frac{1}{p_{i,j}^{(m_2)}},$$

where the first term is the expected time to reach state i from and hence j is positive recurrent.

If there is no positive recurrent state, let i be a null recurrent state, then we will show that all states are null recurrent. For any state j, since the Markov chain is

[1] If P is not diagonizable, we can write $x_0 = \sum_i \alpha_i v_i$ where v_i is a generalized eigenvector. By Theorem 4.7 we still have that $v_1 = \mu$, and $\lambda_1 = 1 > |\lambda_i|$ for $i \geq 2$. Noting that for any Jordan block J with eigenvalue $|\lambda_i| < 1$ we have $\lim_{\mathtt{t}\to\infty} J^{\mathtt{t}} = 0$, ~~and~~ we may obtain a similar result: $\lim_{\mathtt{t}\to\infty} \Pr[X_{\mathtt{t}} = j] = \lambda_1 \mu_j$.

irreducible, we have for some $m_1, m_2 \geq 0$ that $p_{j,i}^{(m_1)}, p_{i,j}^{(m_2)} > 0$. This implies that $\sum_{m=0}^{\infty} p_{j,j}^{(m)} = \infty$ is at least $\sum_{m=0}^{\infty} p_{j,i}^{(m_1)}\, p_{i,i}^{(m)}\, p_{i,j}^{(m_2)} = \infty$, since we have $\sum_{m=0}^{\infty} p_{i,i}^{(m)} = \infty$, since i is a recurrent state. This implies that j is a recurrent state. Since there are no positive recurrent states, it has to be that j is a null recurrent state.

If there are no positive or null recurrent states, then all states are transient. $\square$

4.3.1 Reversible Markov Chains

Suppose there exists a probability vector $\mu = (\mu_i)$ so that

$$\mu_i p_{i,j} = \mu_j p_{j,i}, \qquad i, j \in X. \tag{4.2}$$

Then it is easy to verify by direct summation that μ is an invariant distribution for the Markov chain defined by $(p_{i,j})$. This follows because $\sum_i \mu_i p_{i,j} = \sum_i \mu_j p_{j,i} = \mu_j \sum_i p_{j,i} = \mu_j$, where we used $\sum_i p_{j,i} = 1$. The equations (4.2) are called the *detailed balance equations*. A Markov chain that satisfies these equations is called *reversible*.

Example 4.4 (**Discrete-time queue**)**.** *Consider a discrete-time queue, with queue length $X_{\mathtt{t}} \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$. At time $\mathtt{t}$, $A_{\mathtt{t}}$ new jobs arrive, and then up to $S_{\mathtt{t}}$ jobs can be served, so that*

$$X_{\mathtt{t}+1} = (X_{\mathtt{t}} + A_{\mathtt{t}} - S_{\mathtt{t}})^+.$$

Suppose that $(S_{\mathtt{t}})$ is a sequence of i.i.d. RVs, and similarly $(A_{\mathtt{t}})$ is a sequence of i.i.d. RVs, with $(S_{\mathtt{t}})$, $(A_{\mathtt{t}})$ and X_0 mutually independent. It may then be seen that $(X_{\mathtt{t}},\ \mathtt{t} \geq 0)$ is a Markov chain. Suppose further that each $S_{\mathtt{t}}$ is a Bernoulli RV with parameter q, namely $P(S_{\mathtt{t}} = 1) = q$, $P(S_{\mathtt{t}} = 0) = 1 - q$. Similarly, let $A_{\mathtt{t}}$ be a Bernoulli RV with parameter p. Then

$$p_{i,j} = \begin{cases} p(1-q) & : \ j = i+1 \\ (1-p)(1-q) + pq & : \ j = i, \ \ i > 0 \\ (1-p)q & : \ j = i-1, \ \ i > 0 \\ (1-p) + pq & : \ j = i = 0 \\ 0 & : \ \text{otherwise} \end{cases}$$

Denote $\lambda = p(1-q)$, $\eta = (1-p)q$, and $\rho = \lambda/\eta$. The detailed balance equations for this case are:

$$\mu_i p_{i,i+1} = \mu_i \lambda = \mu_{i+1} \eta = \mu_{i+1} p_{i+1,i}, \qquad \forall i \geq 0$$

These equations have a solution with $\sum_i \mu_i = 1$ if and only if $\rho < 1$. The solution is $\mu_i = \mu_0 \rho^i$, with $\mu_0 = 1 - \rho$. This is therefore the stationary distribution of this queue.

4.3.2 Mixing Time

The mixing time measures how fast the Markov chain converges to the steady state distribution [96]. We first define the *Total Variation* distance between distributions D_1 and D_2 as:

$$\|D_1 - D_2\|_{TV} = \max_{B \subseteq X}\{|D_1(B) - D_2(B)|\} = \frac{1}{2}\sum_{x \in X}|D_1(x) - D_2(x)|.$$

The *mixing time* τ is defined as the time to reach a total variation of at most $1/4$ from the initial state:

$$\|s_0 P^\tau - \mu\|_{TV} = \|p^{(\tau)} - \mu\|_{TV} \leq \frac{1}{4}\|s_0 - \mu\|_{TV},$$

where μ is the steady state distribution and $p^{(\tau)}$ is the state distribution after τ steps starting with an initial state distribution s_0.

Note that after 2τ time steps we have

$$\|s_0 P^{2\tau} - \mu\|_{TV} = \|p^{(\tau)} P^\tau - \mu\|_{TV} \leq \frac{1}{4}\|p^{(\tau)} - \mu\|_{TV} \leq \frac{1}{4^2}\|s_0 - \mu\|_{TV}.$$

In general, after $k\tau$ time steps we have

$$\|s_0 P^{k\tau} - \mu\|_{TV} = \|p^{((k-1)\tau)} P^\tau - \mu\|_{TV} \leq \frac{1}{4}\|p^{((k-1)\tau)} - \mu\|_{TV} \leq \frac{1}{4^k}\|s_0 - \mu\|_{TV}.$$

where the formal proof follows by induction on $k \geq 1$.